37 An explosion in a nuclear reactor spreads the isotope caesium-137, $^{137}_{55}Cs$, across a large area.

After 90.0 years have passed, the quantity of caesium-137 present is 12.5% of its original level.

What is the half-life of caesium-137?

A 11.3 years **B** 22.5 years **C** 30.0 years **D** 45.0 years